

VTU Belagavi

REGULAR EXAMINATION ? 2025

Digital Image Processing ? Regular Examination, 2025

Subject Code: EC603 Semester: 6 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Image Transforms)

1. Describe the working principle of Image Transforms with necessary equations. [12 Marks]
2. Derive the expression related to Image Transforms and explain each term. [12 Marks]
3. List the advantages and limitations of Image Transforms. Explain any two in detail. [8 Marks]

SECTION B (Unit 2: Image Enhancement)

4. Compare and contrast the key concepts involved in Image Enhancement. [12 Marks]
5. How is Image Enhancement implemented? Give an algorithm / procedure and explain. [12 Marks]
6. Compare and contrast the key concepts involved in Image Enhancement. [12 Marks]

SECTION C (Unit 3: Image Restoration)

7. Solve a problem related to Image Restoration. Show all the steps clearly. [12 Marks]
8. Discuss the engineering applications of Image Restoration. [12 Marks]
9. Describe the working principle of Image Restoration with necessary equations. [8 Marks]

SECTION D (Unit 4: Image Segmentation)

10. Write short notes on Image Segmentation. [8 Marks]
11. Compare and contrast the key concepts involved in Image Segmentation. [12 Marks]
12. Describe the working principle of Image Segmentation with necessary equations. [12 Marks]

SECTION E (Unit 5: Image Compression)

13. Explain Image Compression with a neat diagram/sketch and a suitable example. [12 Marks]
14. Describe the working principle of Image Compression with necessary equations. [12 Marks]
15. Compare and contrast the key concepts involved in Image Compression. [12 Marks]

***** End of Question Paper *****